

## Multiple Choice (10 marks)

1. Which of the following sets of bit strings CANNOT be described with a regular expression?
  - (a) All bit strings whose number of zeros is a multiple of five
  - (b) All bit strings starting with a zero and ending with a one
  - (c) All bit strings with an even number of zeros
  - (d) All bit strings with more ones than zeros
2. Which of the following is NOT a reasonable justification for choosing to busy-wait on an asynchronous event?
  - (a) The wait is expected to be short.
  - (b) A busy-wait loop is easier to code than an interrupt handler.
  - (c) There is no other work for the processor to do.
  - (d) The program executes on a time-sharing system.
3. Of the following, which gives the best upper bound for the value of  $f(N)$  where  $f$  is a solution to the recurrence  $f(N) = 2f(N/2) + N$ ?
  - (a)  $O(\log N)$
  - (b)  $O(N \log N)$
  - (c)  $O(\log N) + O(1)$
  - (d)  $O((\log N)^2)$
4. In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol (ARP)?
  - (a) To translate Web addresses to host names
  - (b) To determine the IP address of a given host name
  - (c) To determine the hardware address of a given host name
  - (d) To determine the hardware address of a given IP address
5. Which of the following is true of interrupts?
  - (a) They are generated when memory cycles are "stolen".
  - (b) They are used in place of data channels.
  - (c) They can indicate completion of an I/O operation.
  - (d) They cannot be generated by arithmetic operations.

## True / False (10 marks)

*Answer True or False*

6. True or False: The number of distinct functions mapping a finite set A with m elements into a finite set B with n elements is  $m^n$ .
7. True or False: Reference counting requires additional space overhead for each memory cell to maintain a count of references to it.
8. True or False: The symbol table is a data structure in a compiler that manages information about the identifiers in the source program.
9. True or False: Finding a shortest cycle in an undirected graph G can be solved in polynomial time.
10. True or False: A heap area is usually not represented in a subroutine's activation record frame.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to find numbers divisible by m or n from a list of numbers using lambda function.
12. Write a function to check whether the given number is undulating or not.

*End of Paper*